

Industrial Internship Report on "AI-Based Predictive Maintenance for Industrial IoT"

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Executive Summary

This report details the Industrial Internship provided by upSkill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship focused on completing the Free Summer Internship In Python - 2026 program. Over the course of the program, I completed learning modules and practical assignments, culminating in a final project submission.

My project, "AI-Based Predictive Maintenance for Industrial IoT," was chosen to align with UCT's real-world applications. This internship provided an excellent opportunity to gain exposure to industrial problems and to design and implement a structured, research-oriented Python solution. It was an overall great learning experience.

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1. Preface

This report summarizes the work completed during the 6-week Python internship program. Engaging with practical problem statements is crucial for bridging fundamental concepts with advanced research and implementation.

The opportunity provided by USC/UCT allowed for structured, self-paced upskilling. The program progressed from foundational instructions to hands-on Python coding, ultimately leading to a final project submission on GitHub.

I would like to extend my gratitude to the mentors at upSkill Campus and The IoT Academy. To my peers: dedicating consistent, daily time to foundational programming is the key to building the expertise required for deep research and robust problem-solving.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT), established in 2013, operates in the Digital Transformation domain, providing industrial solutions with a focus on sustainability and RoI. UCT leverages cutting-edge technologies, including the Internet of Things (IoT), Cyber Security, Cloud computing, Machine Learning, Java Full Stack, and Python.

UCT offers several core platforms and solutions:

- * UCT Insight: An IoT platform designed for quick deployment of IoT applications that provides valuable business insights. It features custom dashboards, analytics, and rule engines, supporting industry-standard protocols like MQTT and OPC UA.
- * Factory Watch: A smart factory platform providing scalable solutions for production monitoring, Overall Equipment Effectiveness (OEE), and predictive maintenance, capable of scaling to digital twins.
- * LoRaWAN Solutions: UCT is an early adopter of LoRaWAN technology, delivering solutions across Agritech, Smart Cities, and Industrial Monitoring.
- * Predictive Maintenance: UCT provides industrial machine health monitoring and predictive maintenance solutions by leveraging embedded systems, IoT, and Machine Learning to determine the remaining useful lifetime of production machines.

2.2 About upskill Campus

upSkill Campus (USC), along with The IoT Academy and in association with UCT, facilitated this internship. USC is a career development platform delivering personalized executive coaching in a scalable way. USC aims to upskill one million learners in the next five years by offering self-paced learning alongside support services like internships and career growth services.

The IoT Academy is the EdTech Division of UCT, running executive certification programs in collaboration with EICT Academy, IITK, IITR, and IITG.

2.3 Objective

The objectives for this internship program were to:

- * Get practical experience of working in the industry.
- * Solve real-world problems.

- * Have improved job prospects.
- * Have an improved understanding of our field and its applications.
- * Achieve personal growth, including better communication and problem-solving skills.

3. Problem Statement

Industrial manufacturing relies heavily on the continuous operation of machinery. Unexpected equipment failures lead to severe production bottlenecks, financial losses, and safety hazards. The assigned problem statement focuses on developing a Python-based Machine Learning model to analyze sensor data from industrial equipment to predict potential failures and estimate the Remaining Useful Life (RUL) of the machinery.

4. Existing and Proposed Solution

Existing Solutions and Limitations:

Many current industrial setups rely on reactive maintenance (fixing equipment only after it breaks) or preventive maintenance (scheduled maintenance regardless of actual machine health). Reactive maintenance leads to high downtime, while scheduled preventive maintenance often results in replacing perfectly healthy parts, wasting resources.

Proposed Solution:

The proposed solution is a Python-driven predictive maintenance model utilizing machine learning libraries (such as Scikit-Learn and Pandas). By ingesting historical vibration, temperature, and operational data, the model predicts the likelihood of failure before it occurs, transitioning the maintenance strategy from "preventive" to "predictive". It also includes an interactive web dashboard for real-time monitoring.

Value Addition:

This approach minimizes unexpected downtime, maximizes the lifespan of machine components, and establishes a strong data pipeline that can be easily upgraded to more advanced deep learning frameworks in the future.

Submission Links:

- * Code submission (Github link):

<https://github.com/mannadebraj990-dot/AI-Predictive-Maintenance-IoT>

- * Report submission (Github link):

https://github.com/mannadebraj990-dot/AI-Predictive-Maintenance-IoT/blob/main/Debraj_manna_Internship_Report%20-%20Google%20Docs.pdf

- * Live Web Dashboard: <http://localhost:8501/>

5. Proposed Design/ Model

The system is designed as a streamlined data processing and inference pipeline.

High-Level Design:

1. Data Ingestion: Python scripts read historical telemetry data (e.g., IoT sensor logs) formatted in CSV/JSON.
2. Feature Engineering: Data is cleaned, scaled, and transformed to highlight critical degradation patterns.
3. Model Training & Prediction: A regression-based machine learning model calculates the continuous RUL of the asset based on the extracted features.
4. User Interface: A Streamlit-based web application provides real-time alerts and visualizations (Gauge charts, trends).

Low-Level Design & Interfaces:

- * The core logic is implemented in Python.
- * The system accepts matrix inputs representing multi-sensor time-series data.
- * The output is an expected failure timeline, which can logically interface with a dashboard like UCT's "Factory Watch".

6. Performance Test

Constraints Identified:

- * Accuracy: The model must minimize false positives (predicting a failure when none exists) to avoid unnecessary maintenance checks.
- * Data Processing Speed: The feature extraction scripts must execute efficiently on standard hardware without excessive computational overhead.

Test Plan/Test Cases:

- * Test Case 1: Feed the model a historical dataset of a machine that is known to be healthy. Verify that the predicted RUL remains high and stable (Healthy Status).
- * Test Case 2: Feed the model data showing a gradual spike in vibration and temperature. Verify that the RUL sharply decreases, triggering a maintenance warning (Critical Status).

Performance Outcome:

The algorithm was successfully tested against sample datasets. The Python implementation proved highly efficient for preprocessing large numerical arrays, successfully identifying degradation trends with an acceptable margin of error.

7. My Learnings

This Python internship served as an excellent foundational stepping stone. Building this project reinforced my understanding of applied algorithmic logic and data manipulation.

- * Conceptual Deepening: Transitioning from theoretical syntax to solving a concrete industrial problem solidified my analytical approach.

- * Code Structuring: I learned how to structure Python projects for readability and modularity, which is essential for future collaborative environments.

- * Practical Execution: Submitting the final project, managing version control via GitHub, and building an interactive dashboard has prepared me for more rigorous academic and technical workflows.

8. Future Work Scope

The current implementation provides a robust baseline for predictive modeling. Future work will focus on integrating more sophisticated Deep Learning and Neural Network architectures to handle complex, non-linear sensor data. Additionally, natural language processing (NLP) could be integrated to automatically generate readable maintenance reports from the raw analytical data, creating a more comprehensive AI-driven industrial assistant.